

Jixin Han

TECH LEAD, SENIOR AI ENGINEER AND RESEARCHER

CONTACT

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EDUCATION

Ph.D. / Computer Science and Engineering
Waseda University
2017~2022

Research interests: C/C++ compiler optimization, mathematical verification, and formal verification

Master / Computer Science and Engineering
Waseda University
2014~2017

Research interests: high performance computing, C/C++ compiler optimization

Bachelor / Computer Science and Engineering
Beijing Language and Culture University

LANGUAGE

- English: TOEFL 113/120
- Japanese: N1, Native
- Chinese: Native

PROFILE

Tech Lead and Senior Autonomous Driving Engineer focused on model prototyping, including sensor fusion, temporal fusion, and end-to-end (E2E) systems. Specialized in auto-labeling pipelines, developing data validation and sampling tools, and enhancing label quality through active learning and ensemble methods. Focused on optimizing inference for edge devices using TensorRT, HailoRT, custom CUDA operations, pruning, and quantization to achieve significant speedups. Also delivered advanced online courses on CUDA and TensorRT in China. Received Ph.D. from Waseda University in 2022.

PROFESSIONAL EXPERIENCE

TECHNICAL LEAD

TIER-IV | Tokyo, Japan | 2024.07~Current

Specialized in model prototyping, deployment, integration, auto-labeling, and MLOps for autonomous driving. Using C++, CUDA, Python, Go.

- Prototyped multi-task models using sensor fusion, active learning, temporal fusion, and shrink the model size while keeping precision.
- Built auto-labeling system, and generated 2D/3D detections, panoptic segmentation, lane detection. Improved precision using ensembling.
- Deployed and integrated models on edge devices using C/C++, CUDA, TensorRT and HailoRT, applied quantization and created custom toolkits to analyze and optimize models.
- Built MLOps pipelines with Kubernetes and Kubeflow for automated data workflows and CI/CD/CT on GPU clusters

TECHNICAL LEAD

T2 | Tokyo, Japan | 2024.01~2024.06

Led the perception team at T2, a logistics-focused startup, driving model prototyping, deployment, and system integration. Using C++, CUDA, Python.

- Developed a high-precision BEVFusion model and customized the model structure to fully exploit hardware usage.
- Optimized inference speed with TensorRT and CUDA, pruning, quantization, and designed customized operations to boost speed.
- Developed a set of C++ benchmarking tools to evaluate training and deployment performance
- Built sensor synchronization algorithm ensuring precise multi-sensor data fusion.

ONLINE WEBINAR LECTURER

Udemy China | Shanghai, China | 2023.01~2025.06

Taught an online course in China on a platform similar to Udemy, covering high-performance deployment with CUDA and TensorRT for autonomous

PROGRAMMING & FRAMEWORKS

- C/C++
- Python
- Go
- NodeJS
- Darknet
- PyTorch
- ONNX
- CUDA/cuDNN/TensorRT
- protobuf

OTHERS

- Apollo
- ROS2
- Kubernetes (k8s)
- Kubeflow

- Deployed models ranging from image-only and lidar-only, to sensor fusion models.
- Explained advanced computer architecture and parallel processing concept using C++ and CUDA
- Demonstrated strategies to preserve model precision during deployment from PyTorch to ONNX and TensorRT

ENGINEER AND RESEARCHER

Honda Motor Co., Ltd. | Tokyo, Japan | 2022.04~2023.12

Developed multi-task DNNs for autonomous driving—including detection, segmentation, keypoints, optical flow, and tracking—applying active learning and pseudo-labeling to boost performance on limited data.

- Developed multi-task DNNs for detection, segmentation, keypoints, optical flow and tracking.
- Applied active learning and pseudo-labeling to enhance performance on limited datasets
- Optimized in-vehicle deployment with C++ parallelism, custom CUDA ops, TensorRT, pruning, and quantization.

PRESIDENT

IEEE HKN Mu-Tau Chapter | Tokyo, Japan | 2019.04~2020.09

Served as President of Japan's first IEEE-HKN chapter at Waseda University, leading international collaborations with U.S. and European chapters.

- Organized joint academic events and exchange program.
- Guided the chapter to win the IEEE-HKN Outstanding Chapter Award in Boston in 2019.
- Fostered leadership and networking opportunities across global IEEE student communities.

ACADEMIC DISCIPLINES

- (First) "Durable Queue Implementation built on Formally Defined Strand Persistency Model", Journal of Information and Processing, Jun 2021,
- (First)"Parallelizing Compiler Translation Validation Using Happens-Before and Task-Set", CSA'21, Oct 2021
- (Co-author) "Fast and Highly Optimizing Separate Compilation for Automatic Parallelization", HPCS'19, Jul 2019
- (First) "Reducing parallelizing compilation time by removing redundant analysis", SPLASH'16, Oct 2016
- (First) "Proving the Correctness of NVM Recovery under Stand Persistency Model", IPSJ-PRO, Sep 2021
- (First) "The Translation Validation for the Coarse-grain Parallelizing Compiler", IPSJ-PRO, Sep 2020
- (First) "Verified Translation Validation Technique for OSCAR Automatically Parallelizing Compiler", IPSJ-PRO, Sep 2018
- (Co-author) "Code Generating Method with Profile Feedback for Reducing Compilation Time of Automatic Parallelizing Compiler", Mar 2017